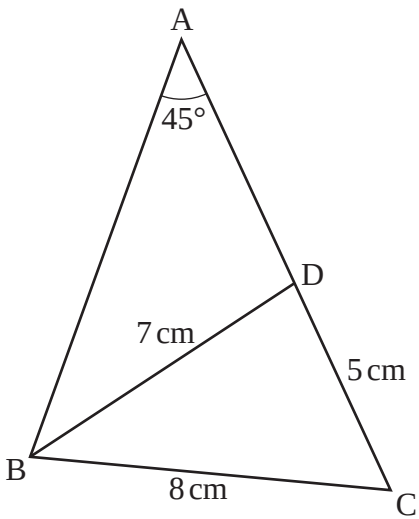
11 In this question you must show detailed reasoning.

The diagram shows triangle ABC, with BC = 8 cm and angle BAC = 45°.

The point D on AC is such that DC = 5 cm and BD = 7 cm.



Determine the exact length of AB. **[5]**